

01 SELECT REFERENCE + GRAPH

GRCh38 / HPRC1

GRCh38 / CPC1 + HPRC1

CHM13 / CPC2*

Same choice governs
graph and coordinates

02 CHOOSE AN INPUT AND PLATFORM

NGS · Illumina

FASTQ

R1 + R2

Personalized graph

Graph alignment

BAM checks

Mutect2

small

SNV

+ Indel

NGS: normalize + dedup

Direct entry →

BAM + BAI

HiFi · PacBio

FASTQ

HiFi reads

Personalized graph

Graph alignment

BAM checks

HiFi: normalize + index

HiFi: small / sv / all

DeepSomatic

small

SNV

+ Indel

Sniffles2

sv

SV

VCF + SNF

GRAPH SOURCE

Paired: normal → shared graph* | Tumor-only: tumor → graph

full

FASTQ → BAM checks → callers

align_only

FASTQ → BAM checks → stop

call_from_bam

BAM → checks → callers

* CPC2 not public. Paired FASTQ: known issue (see audit). Paired SV: separate calls, no subtraction.